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Increasingly sophisticated image editing tools make it easy to modify images. Often these modifications are elaborate, convincing, and undetectable by even careful human inspection. These considerations have prompted the development of forensic algorithms and approaches to detect modifications done to an image. However, these detectors are model-driven (i.e., manipulation-specific) and the choice of a potent detector requires knowledge of the type of manipulation, something that cannot be known (*a priori*). Thus, the latest effort is directed towards developing model-free (i.e., generalized) detectors capable of detecting multiple manipulation types. In this article, we propose a novel detector capable of exposing seven different manipulation types in low-resolution compressed images. Our proposed approach is based on a two-layer convolutional neural network (CNN) to extract frequency domain features of image median filtered residual that are classified using two different classifiers—softmax and extremely randomized trees. Extensive experiments demonstrate the efficacy of proposed detector over existing state-of-the-art detectors.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Machine learning approaches**;

Additional Key Words and Phrases: Multiple manipulation detection, image forensics, Convolutional neural network (CNN), two-layer architecture

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1 INTRODUCTION

Images are authenticated using digital image forensic methods that uncover tampering traces based on the statistical attributes of tampered images or based on some initial information about the original images. However, statistical attributes of forged images strongly depend upon the type of manipulation used to tamper the image. Consequently, image forensic methods are specific to detect particular manipulation types [48]. Such forensic methods are designed for re-sampling

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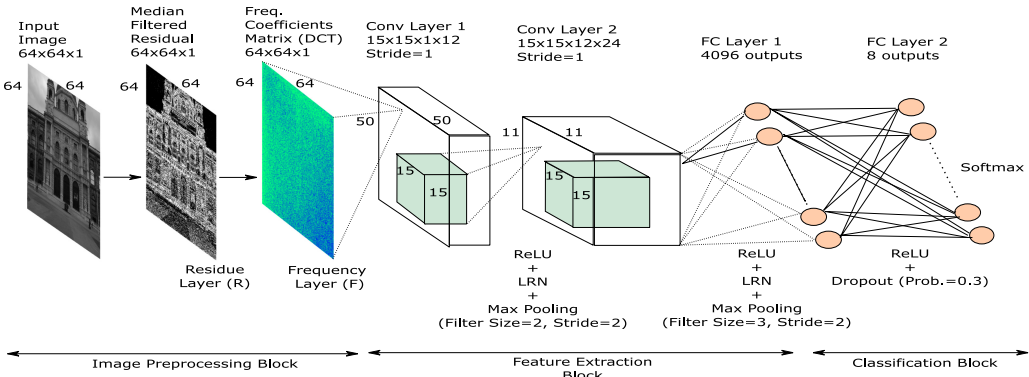


Fig. 1. The RF-CNN network architecture. Acronyms ReLU and LRN are used for rectified linear unit and local response normalization.

detection [17, 40], quantization step estimation [52, 56], median filtering detection [9, 10, 20, 28, 29, 51, 55, 59], and contrast enhancement detection [7, 47, 57].

Dedicated forensic methods are limited in determining the processing history of an image. A forensic analyzer needs to test the questionable image with multiple dedicated detectors to ensure image authenticity. Even so, because of the reject-class problem, i.e., a detector does not know when it is not an appropriate detector, it becomes yet harder to accommodate the results of multiple detectors into a reliable decision. Also, anti-forensic methods can easily be designed to fool such specific detectors. Furthermore, a tampered image usually goes through various operations for the tampering to be visually imperceptible and it is required to detect the sequence of all the operations performed on the image. To address these issues, recent focus has shifted towards designing detectors that are model-free and can detect multiple image manipulations [5, 8, 16, 34, 41, 50]. However, these general purpose detectors are designed using hand-crafted features [16, 34, 41] and convolutional neural network (CNN) [31] based features [5, 8, 12, 50].

Most of the passive image processing operations (image filtering, image re-scaling, and image rotation) significantly change the frequency content of an image. In this context, we propose a novel CNN-based detector by incorporating two new pre-processing layers—residue (R) and frequency (F). We use the frequency domain representation of image median filtered residual as input to a CNN. Hertel et al. [23] have experimentally shown that in a deep learning framework, more discriminative features for an input audio signal can be learned from frequency domain input in comparison to time domain input. This gives us a basis for using image median filtered residual in the frequency domain as input to a CNN. The features evolved by the CNN are classified using two different classifiers—softmax [6] and extremely randomized trees (ERT). Our overall architecture consists of three stages: a pre-processing stage, a core CNN stage for extracting higher-level features, and a classification stage, as shown in Figure 1. We refer to our overall architecture as RF-CNN.

RF-CNN can detect seven different image manipulation types (average filtering, Gaussian filtering, median filtering, Wiener filtering, Laplacian filtering, rescaling, and rotation) in low-resolution and JPEG compressed images. The results of RF-CNN are compared with conventional machine learning-based detectors GLF [9] and $SRM_{(T1, O3, q^*)}$ [34]. We have also compared our results with recent CNN-based detectors MFR-CNN [10] and MISLnet [5]. Exhaustive experimentation for both single and multiple manipulation detection demonstrate the robustness of RF-CNN compared to state-of-the-art detectors. RF-CNN is also computationally efficient because of a simple two-layer architecture as compared to other CNN-based methods. Due to computationally efficient

architecture of RF-CNN, it can be easily deployed in mobile computing devices where power usage is always an issue.

Our contributions in this article are threefold. First, we present a technique for manipulation detection in low-resolution compressed images. Second, we present a relatively smaller feature extractor CNN network that results in lower computational and time complexity. Third, as a consequence of a smaller network, we discuss the possibility of using a comparatively smaller amount of training data to train RF-CNN.

The article is organized as follows: Brief literature review and description of conventional CNN architecture are given in Sections 2 and 3, respectively. Section 4 illustrates the image pre-processing layers, and Section 5 provides a more detailed description of the RF-CNN. Section 6 provides details on the methodology used for evaluating the performance of RF-CNN, and Section 7 provides the experimental results. We conclude the article with some observations in Section 8.

2 RELATED WORKS

In recent years, various universal image manipulation detectors have been proposed. Among hand-crafted features-based detectors, Qiu et al. [41] utilized steganalytic models SPAM [39], SRM [18], and LBP [46] to detect the type of image tampering. In another work [16], small patches of an image are modeled by a Gaussian mixture model (GMM) followed by two-class classification to verify the authenticity of images. Li et al. [34] proposed an SRM [18] based reduced feature set for multiple image manipulation detection. Rather than engineering the features followed by empirical estimation (e.g., classification), a promising direction is to let the feature detection be part of the empirical estimation and the features automatically evolve as part of the training process. The hidden layers of a CNN, for example, serve as feature detectors and the features are usually quite robust. CNNs have been successfully applied to image classification [33], document image classification [27], pose estimation [53], action recognition [58], and passive image forensics [5, 8, 10, 12, 51, 57].

Chen et al. [10] utilized median filtered residual (MFR) of an image as input to a CNN to distinguish between median filtered and original images. In Reference [12], residual-based local features are extracted using CNN and utilized for general purpose manipulation detection. In another work [8], high pass filtered images with CNN are used to extract deep features for image processing operation detection. Another CNN-based detector [51] has employed interpolated images as input to CNN to detect median filtering in small resolution images. Further, in Reference [5], the authors proposed a constrained CNN to extract high-level features of differently manipulated (median filtering, Gaussian filtering, rescaling, JPEG compression, and additive white Gaussian noise (AWGN) addition) images to classify among five different manipulation types. In a recent work [60], a two-stream Faster R-CNN network is used to detect the manipulated regions in an input image. However, existing general purpose detectors mainly distinguish among different manipulation types for uncompressed large resolution images. Thus, identification of different image processing operations performed on low-resolution images under the attack of JPEG compression is still to be addressed. Moreover, discriminative features can be learned better by the CNN if the input is in frequency domain [23]. Thus, we use RF-CNN, which can detect seven different manipulation types in low-resolution images with high accuracy.

3 CONVOLUTIONAL NEURAL NETWORKS

CNNs are a specialized form of neural network that are particularly well suited for images [31]. Deep CNN architectures typically alternate convolutional layers followed by non-linear activation function and sub-sampling layers (e.g., pooling layers) with final fully connected layers to provide the overall output.

The convolutional layer performs the convolution of input channels with the sliding window convolution filter where sliding distance is known as stride. Usually, multiple convolution filters with learnable filter weights are used in convolutional layer. Each of the filter weights are convolved with each input channel and summed up to obtain single feature map corresponding to each filter.

For K feature maps of $(l-1)^{th}$ layer, the output corresponding to l th convolutional layer n th learnable filter of size $Z \times Z$ is given by

$$y_n^{(l)} = \sum_{k=1}^K \sum_{i=1}^Z \sum_{j=1}^Z w_n^{(l)}(i, j) \times y_n^{(l-1)}(i, j, k) + b_n^{(l)}, \quad (1)$$

where (i, j) represents the position of the filter window overlapped with feature map window and $b_n^{(l)}$ represents the corresponding bias term. The feature extraction step in CNN is termed as the forward propagation step. The filter weights $w_n^{(l)}$ are initialized with random values and learned using the back-propagation algorithm [13, 32].

The learnable parameters of a CNN (filter weights and bias) along with fully connected layers weights are optimized so that an overall cost function is minimized. The cost function for m training examples and P different class labels is defined as

$$J = -\frac{1}{m} \sum_{i=1}^m \sum_{j=1}^P y_i^j \log(\hat{y}_i^j), \quad (2)$$

where $\hat{y}_i^j$ is the predicted output for i th training image of j th class and y_i^j denotes the corresponding true class label. We update the parameters using stochastic gradient descent (SGD) [42] and minimize J . The updation rule using SGD is given by,

$$\begin{aligned} w_n^{(l)} &= w_n^{(l)} - \alpha \frac{dJ}{dw}, \\ b_n^{(l)} &= b_n^{(l)} - \alpha \frac{dJ}{db}, \end{aligned} \quad (3)$$

where α is the learning rate.

4 PROPOSED IMAGE PRE-PROCESSING LAYERS

4.1 The Residue (R) Layer

Conventionally, CNN is used directly with raw images and has been widely used in various image processing applications such as image classification, object tracking, document classification, action recognition, human pose estimation, and so on, by exploiting the features extracted from raw images. In the field of image forensics, the image is generally pre-processed before feeding to conventional CNN network [5, 8, 10, 51, 57]. Image pre-processing helps to suppress image content and to enhance the effects of manipulations performed on the images. In recent works [5, 10, 34], it has been demonstrated that image residual-based features are more efficient than raw image features to detect image manipulations. Thus, in RF-CNN, a residue layer is placed at the beginning of the network to determine the median filtered residual (MFR) [28] of an image, which is defined as

$$I^{MFR} = mf_w(I) - I, \quad (4)$$

where I^{MFR} is the median filtered residual of image I and mf_w denotes median filtering operation with window size $w \times w$.

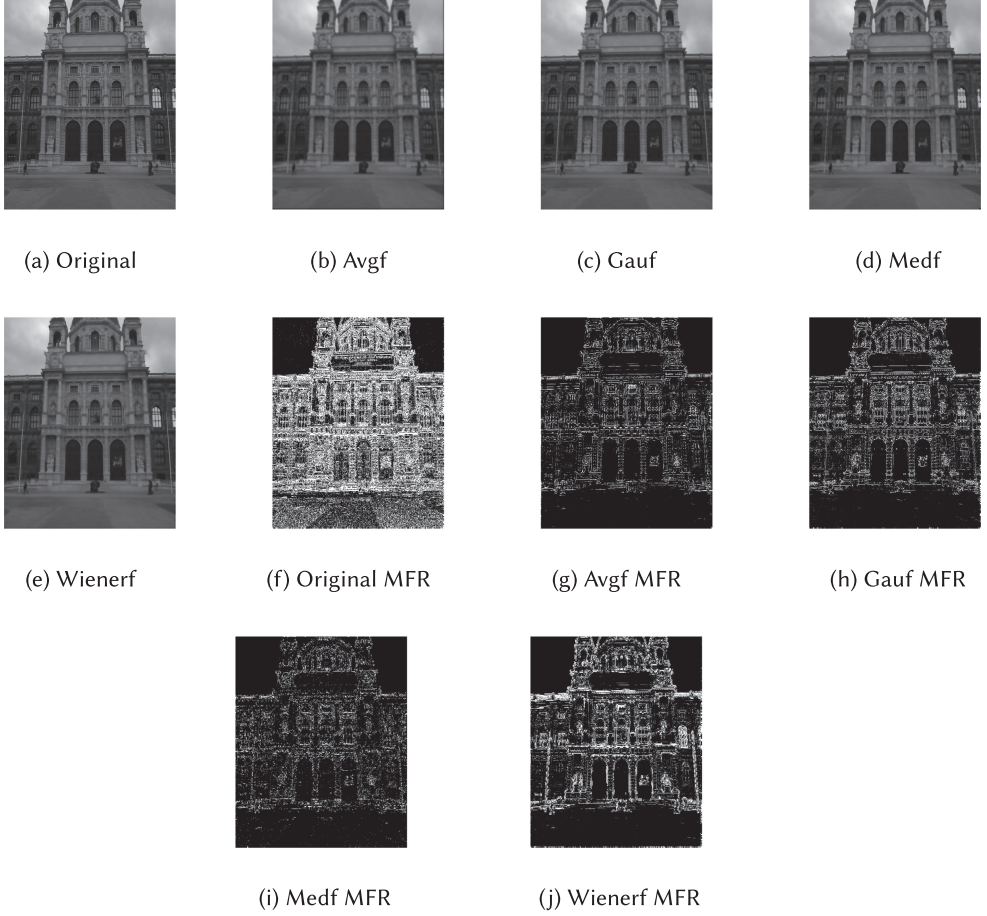


Fig. 2. (a) An original image from the BOSSBase [3] database, (b) Average filtered image (Avgf), (c) Gaussian filtered image (GauF) (standard deviation $\sigma = 1.1$), (d) Median filtered image (Medf), (e) Wiener filtered image (Wienerf), (f)–(j) Corresponding median filtered residuals (MFRs) of Figures 2(a)–2(e). Images in Figures 2(a)–2(e) are filtered with window size 5×5 and MFRs are displayed log compressed. Avgf MFR, GauF MFR, Medf MFR, and Wienerf MFR denote MFRs of corresponding filtered images.

Figure 2 shows the original image, manipulated images using various operations along with MFRs of original and manipulated images. Figure 2 shows that original images and images manipulated using different filtering operations can be clearly distinguished in residual domain (Figures 2(f)–2(j)) rather than spatial domain (Figures 2(a)–2(e)). Thus, MFR of an image can effectively suppress image content and magnify fingerprints of image manipulation.

4.2 The Frequency (F) Layer

Many appearance-preserving image processing operations, i.e., spatial domain image filtering (average filtering, median filtering, Gaussian filtering, Wiener filtering, and Laplacian filtering), up-sampling, down-sampling, JPEG compression, and so on, significantly alter the spatial frequency content of an image while maintaining its visual appearance. Thus, spatial frequency information of images provides clues about the image processing method used for manipulating an image. In this work, spatial frequency information is extracted using the discrete cosine transform

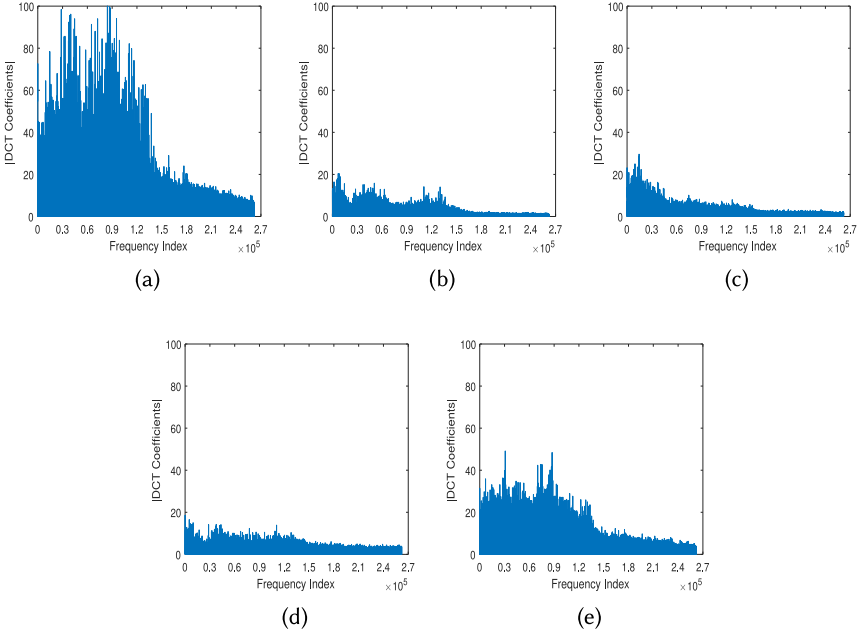


Fig. 3. Plots of DCT coefficients magnitude (scanned in row-major order) against increasing frequency index (a) Original image MFR (Figure 2(f)), (b) Average filtered image (Avgf) MFR (Figure 2(g)), (c) Gaussian filtered image (Gauf) MFR (Figure 2(h)), (d) Median filtered image (Medf) MFR (Figure 2(i)), and (e) Wiener filtered image (Wienerf) MFR (Figure 2(j)).

(DCT) of residue images. The magnitude of DCT coefficients at each location of DCT matrix represent different frequency content in an image MFR. To visualize the frequency content of original and manipulated images MFRs, magnitudes of image MFRs DCT coefficients are arranged in 1-D array by scanning in row-major order (ascending frequency index order) and plotted against increasing frequency index in Figure 3. It should be noted that DCT coefficients are determined for Figures 2(f)–2(j) having size 512×512 and 262,144 DCT coefficient values (frequency index values) are obtained. Figure 3 shows significant reduction in the amplitudes of DCT coefficients of average filtered (Avgf) image MFR, Gaussian filtered (Gauf) image MFR, median filtered (Medf) image MFR, and Wiener filtered (Wienerf) image MFR as compared to DCT coefficients of original image MFR. However, the variation in magnitude of DCT coefficients is shown for particular filtering manipulations (Avgf, Gauf, Medf, Wienerf); the similar effects can also be observed for rescaling, image rotation, and Laplacian filtering operations.

Thus, a frequency (F) layer is added before the proposed network that converts image MFR into frequency domain using DCT. DCT coefficient matrix of the image MFR is further fed into proposed CNN network. Also, conventional CNN requires a deep network architecture to extract efficient features directly from image pixel values. In this work, we have proposed a shallow CNN network along with image pre-processing to take advantage of both conventional machine learning methods (small training time) and deep learning methods (high-level feature extraction) for the purpose of image forensics.

5 PROPOSED NETWORK ARCHITECTURE

The proposed network is mainly classified into three blocks: image pre-processing block, feature extraction block, and classification block, as shown in Figure 1. Image pre-processing block

transforms the input space (from pixel intensity values) to a residual frequency representation to guide the formation of frequency-specific features in the following stage. Appearance-preserving manipulations leave the image indistinguishable from the original image but different in its spectral characteristics. The output of image pre-processing block is fed into feature extraction block. In the proposed network, the feature extraction block is composed of two convolutional layers that extract high-level features of different image manipulations. Finally, the extracted features are fed into classification block for identification of the manipulation type based on the output of the classifier. The detailed description of each layer specification is given below.

5.1 Image Pre-processing Block

The image pre-processing block includes the above-defined residue (R) and frequency (F) layers that accepts a single channel gray scale image and outputs the DCT matrix of image MFR. The size of the output from the image pre-processing block remains the same as the input image. In Figure 1, the proposed network architecture is shown for an input image of size $64 \times 64 \times 1$. It should be noted that in Figure 1, the DCT matrix is log compressed and shown in the form of colormap image. In this work, the network architecture shown in Figure 1 is used for images of size 64×64 and above. However, for images of size 32×32 , experiments are conducted by removing both pooling layers in feature extraction block.

5.2 Feature Extraction Block

The feature extraction block extracts refined features of different image manipulation operations using two convolutional layers. Both of the convolutional layers are followed by an activation function layer, normalization layer, and pooling layer. The higher-level extracted features are further fed into classification block for classification into different image manipulation operations. The specifications of different layers used in feature extraction block are given below.

5.2.1 Convolutional Layers. In the proposed network, two convolutional layers Conv Layer 1 and Conv Layer 2 are used with kernel size 15×15 . Number of filters used in Conv Layer 1 and Conv Layer 2 are 12 and 24, respectively. For an input of size $64 \times 64 \times 1$, 12 different convolutional filters of Conv Layer 1 with a stride of 1 are convolved with the input to generate output feature maps of dimension $50 \times 50 \times 12$. Similarly, Conv Layer 2 with stride 1 inputs the feature maps of dimension $25 \times 25 \times 12$ and outputs feature maps of dimension $11 \times 11 \times 24$. In contrast with state-of-the-art CNN-based forensic methods [5, 8, 10, 51, 57], the proposed network contains only two convolutional layers with relatively larger kernel size to extract global features of an image. Also, the number of filters in each convolutional layer are limited to 12 and 24 to significantly reduce the training time. Experiments have been conducted with various convolutional filter kernel sizes, number of filters in Conv Layer 1 and Conv Layer 2, and results are reported in Figures 4(a) and 4(b), respectively. To this end, a database Φ of 13,985 images is created by combining 1,338 images from each of UCID [44] and BOWS2 [4] databases, 10K images from BOSSBase [3] and 1,309 images from NRCS [37] database. The images are centrally cropped to size 32×32 and 64×64 and post-JPEG compressed with $Q = 70$ to create Φ^S where $S \in \{32 \times 32, 64 \times 64\}$. Then, average accuracies for eight class classification of 32×32 sized images with different filter sizes and number of filters are determined. Figure 4(a) shows maximum accuracy for filter kernel size 15 whereas Figure 4(b) shows that increase in number of filters slightly increase validation accuracy with the expense of large training time. It should be noted that filter kernel size greater than 15 cannot be used for both image sizes 32×32 and 64×64 .

5.2.2 Activation Function. Activation functions are used to introduce non-linearity in the network, i.e., to extract non-linear higher-level feature maps. There are many types of activation functions used in deep learning applications, for, e.g., sigmoidal activation function,

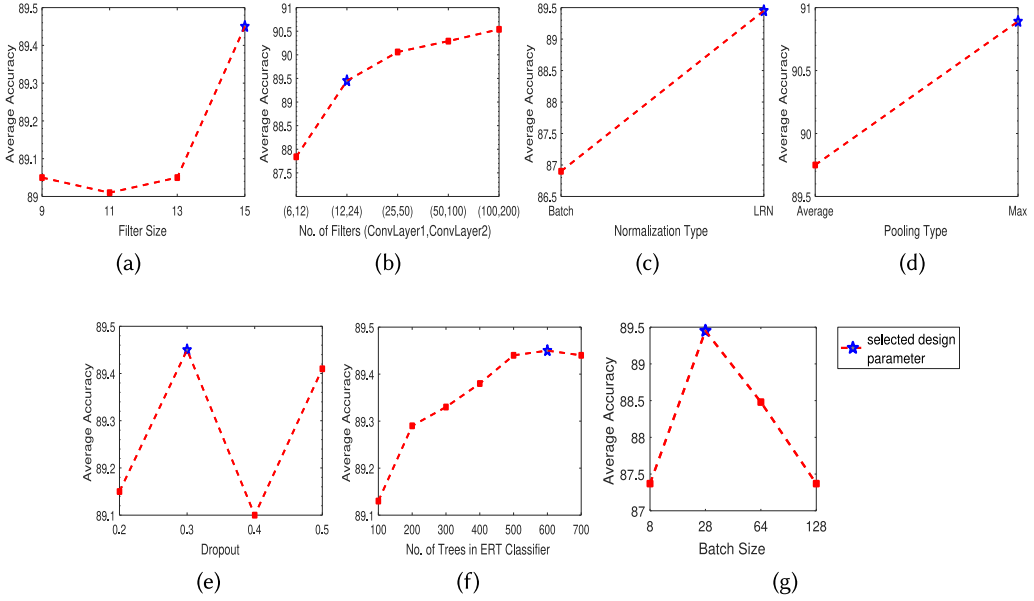


Fig. 4. Experimental selection of RF-CNN design parameters based on eight class average classification accuracies achieved for different: (a) Filter sizes, (b) Number of filters in Conv Layer 1 and Conv Layer 2, (c) Normalization types, (d) Pooling types, (e) Dropout, (f) Number of trees in ERT classifier, and (g) Batch sizes. Blue star-shaped marker represents selected parameter. The eight classification classes include original compressed ($Q = 70$) image class and seven manipulated compressed image classes, images being manipulated with operations listed in Table 1 with blue highlighted parameters.

Hyperbolic tangent ($\tanh$) activation function, and rectified linear unit (ReLU) activation function [35] and its variants [54]. ReLU activation function typically results in faster convergence of the stochastic gradient descent (SGD) and is thus widely used in deep learning applications [30, 49]. Many variants of the standard ReLU activation function such as Leaky ReLU [35], parametric ReLU [22], and exponential linear units (ELU) [11] are being used in recent works. However, leaky ReLU and Parametric ReLU are not robust to noise compared to ReLU [11].

In this work, ReLU [35] activation function is used after each convolutional layer and fully connected layer 1 (FC1) to speed up the training process. However, softmax activation function [6] is used after fully connected layer 2 (FC2). Softmax activation is used as a generalized version of sigmoid activation function for multiclass classification.

5.2.3 Local Response Normalization (LRN). Data normalization is a well-known method to accelerate the training of a conventional machine learning or deep learning model. For CNN-based networks, two methods, local response normalization (LRN) and batch normalization, are successfully used for data normalization. LRN normalizes the data within a single feature map around its neighboring values, whereas batch normalization normalizes data by maintaining the input distribution to a learning network [25]. In RF-CNN, LRN is used after each convolutional layer, which is followed by activation function as shown in Figure 1. Figure 4(c) shows comparison of eight class classification average accuracies for images of $\Phi^{32 \times 32}$ for proposed CNN network using LRN and batch normalization.

5.2.4 Pooling Layer. In deep learning networks, it is always beneficial to aggregate the low-level features over a particular window (overlapping or non-overlapping) to reduce spatial invariance [45]. This phenomenon is known as pooling, which results in reduction of the size of the feature

maps. Two different pooling methods are often used: Average pooling and Max pooling. We first use a max pooling layer with window size 2, a stride of 2; and a second max pooling layer with window size 3, a stride of 2 is used. Both pooling layers are used immediately after normalizing data from each of the LRN layers. First and second pooling layers reduce the size of feature maps from $50 \times 50 \times 12$ to $25 \times 25 \times 12$ and from $11 \times 11 \times 24$ to $5 \times 5 \times 24$, respectively. It should be noted that for image sizes 64×64 and above, both max pooling layers are used, but for image size 32×32 , no pooling layer is used, as feature maps already have very small sizes. A comparison of average and max pooling layers, in terms of eight class classification average accuracies achieved for images of $\Phi^{64 \times 64}$ is given in Figure 4(d).

5.3 Classification Block

The classification block classifies the image tampering operation done on a particular test image by assigning a probability to each image tampering operation class. It mainly consists of fully connected layers along with dropout layer and softmax activation function. Classification is performed using two different classifiers—softmax classification layer and extremely randomized trees (ERT) classifier.

We use a fully connected layer (FC1) with 4,096 neurons to extract higher-level features association from features which are extracted from feature extraction block. Another fully connected classification layer (FC2) with eight neurons is used to classify among eight image tampering classes. Dropout layer is used to reduce overfitting problems in deep neural networks [24] by deselecting some of input values and generate classification results based on remaining feature set. A dropout layer after fully connected layer (FC1) is used with probability 0.3 (Figure 4(e)), which randomly consider 70% output features from FC1 to determine class probabilities. Figure 4(e) shows comparison of different dropout values in terms of eight class classification average accuracies for images of $\Phi^{32 \times 32}$.

In RF-CNN, the features extracted from fully connected layer FC1 (4,096 features) are also fed into ERT classifiers for classification. ERT classifier works by fitting randomized decision trees on subsets of training features and then averaging the performance. The use of ERT classifiers for classification is inspired by its better performance as shown in Reference [5]. We perform classification with 100, 200, . . . , 700 randomized decision trees and achieved maximum average validation accuracy with 600 decision trees (Figure 4(f)). Thus, 600 randomized trees are used for further experimentation.

5.4 Differences from Existing CNN Architectures

Existing CNN architectures for digital image forensics [5, 8, 10, 51, 57] make use of a deep architecture for image feature extraction. In RF-CNN, image pre-processing is done to determine more efficient lower-level features before extracting higher-level features from feature extraction block. In contrast to existing methods, our proposed network contains only two convolutional layers with larger kernel size (15×15). As the proposed network is specifically designed for low-resolution images of size 32×32 and 64×64 , the proposed convolutional layers can efficiently extract global features of images with significant reduction in training time of the proposed model. Along with this, only two fully connected layers are used in the proposed network in contrast to three fully connected layers in existing networks [5, 10, 57].

6 EXPERIMENTAL SETUP

6.1 Image Database

For detailed evaluation of RF-CNN, database Φ of 13,985 images has been used with various image resolutions. RF-CNN results are evaluated for image sizes 256×256 , 128×128 , 64×64 , and

Table 1. Image Manipulation Types with Editing Parameters

Manipulation type	Parameters
Average filtering (Avgf)	$w_{size} = 3, 5$
Gaussian filtering (Gauf)	$\sigma = 1.1, w_{size} = 3, 5$
Laplacian filtering (Lapf)	$\alpha = 0.2$
Median filtering (Medf)	$w_{size} = 3, 5$
Rescaling operation (Res) using bilinear interpolation	Scale = 1.2, 1.5
Rotation operation (Rot) followed by cropping to same size	$1^\circ, 3^\circ$
Wiener filtering (Wienerf)	$w_{size} = 3, 5$

32×32 . To this end, images of database Φ are centrally cropped to size 256×256 , 128×128 , 64×64 , and 32×32 to create a new database Φ^S where $S \in \{256, 128, 64, 32\}$. Further, the images of dataset Φ^S for $S = 32, 64$ are manipulated using different manipulations with parameters as given in Table 1 and JPEG compressed with quality factor $Q = 70, 30$. However, for $S = 128, 256$, images are manipulated with operations listed in Table 1 with only one parameter (average filtering, median filtering, Gaussian filtering and Wiener filtering with window size $w_{size} = 5$, Laplacian filtering with shape parameter $\alpha = 0.2$, rescaling with factor 1.5, and image rotation with 3° followed by cropping to same size) and JPEG compressed with compression quality factor $Q = 70$. For simulation of GLF [9], SRM_(T1, O3, q*) [34], and MFR-CNN [10] results, the same database Φ^S for $S = 32$ and 64 is used as databases in respective manuscripts of comparable size.

However, the database used to evaluate MISLnet performance in Reference [5] is relatively larger than Φ^S . Thus, for simulation of MISLnet [5] results, the database is created by randomly selected 1,013 images from IEEE IFS-TC image forensics challenge [14]. The tiles of size 64×64 are cropped from high-resolution images (1024×768) and central 140 tiles per image are selected from 192 tiles. As mentioned in Reference [5], training dataset for single manipulation detection is created by randomly selected 25K original and corresponding 25K manipulated (manipulated with operations listed in Table 1) images, whereas test dataset is created by randomly selected 5K original images along with 5K corresponding manipulated images (manipulated with operations listed in Table 1). Similarly, a training dataset of 133,336 images (16,667 images correspond to original and each of the seven operations) is created for multiple manipulation detection. Test dataset for multiple manipulation detection is created by combining 5,337 images of original and 5,337 images of each of seven different manipulation types. For multiple manipulation detection, images are manipulated with operations listed in Table 1 with only one parameter (average filtering, median filtering, Gaussian filtering and Wiener filtering with window size $w_{size} = 5$, Laplacian filtering with shape parameter $\alpha = 0.2$, rescaling with factor 1.5, and image rotation with 3° followed by cropping to same size). All datasets are compressed with quality factor $Q = 70$ before classification.

6.2 Experimental Methodology

The proposed CNN network is specifically designed to detect passive image manipulation operations in small images, which is still a challenge in the field of image forensics. To address this challenge, performance of proposed CNN network is evaluated with exhaustive experiments and compared with recent conventional machine learning detectors GLF [9], SRM_(T1, O3, q*) [34] and CNN-based detectors MFR-CNN [10], MISLnet [5]. The proposed network is designed to

identify seven different image manipulation types. Image manipulation types along with manipulation parameters used are tabulated in Table 1.

For simulation of results of hand-crafted features-based detectors GLF [9] and $SRM_{(T1, O3, q*)}$ [34], classifier settings are used as given in respective manuscripts and experiments are conducted using MATLAB 2017b. For proposed RF-CNN and MFR-CNN [10] networks, all the experiments on Φ^S images are conducted using MATLAB 2017b with Nvidia Quadro K-420 single GPU system with 8 GB RAM. Also, features are extracted from fully connected layer FC1 of RF-CNN using MATLAB and further used to train ERT classifier using Caffe [26] deep learning framework. It should be noted that 600 randomized trees are used for single and multiple manipulation detection. For compilation of MISLnet [5] results, Caffe deep learning framework [26] with Python version 3.6 is used. Settings for training MISLnet [5] are used as mentioned in respective manuscript. It is worth mentioning here that MISLnet [5] results are simulated for image resolution 64×64 only as original network requires modification for images of smaller resolutions.

For RF-CNN and MFR-CNN [10], for a database of 13,985 original images, the number of training epochs is set to 10 (approximately 6,300 iterations for two-class classification and 29,790 iterations for eight class classification) where one epoch means going through the complete database once. The batch size is set to 28 based on experimentation with different batch sizes (Figure 4(g)). The results are determined in terms of eight class classification average accuracy for images of size 32×32 .

7 RESULTS

7.1 Single Manipulation Detection

To evaluate the performance of RF-CNN to detect image manipulations listed in Table 1, results are simulated for small images (image sizes 64×64 and 32×32) compressed with quality factor Q ($Q = 70, 30$). RF-CNN's results are compared with conventional hand-crafted features-based detectors GLF [9], $SRM_{(T1, O3, q*)}$ [34] and CNN-based detectors MFR-CNN [10], MISLnet [5] and shown in Table 2. It should be noted that detectors RF-CNN, GLF [9], $SRM_{(T1, O3, q*)}$ [34], and MFR-CNN [10] results are simulated for both image resolutions 32×32 and 64×64 . However, detector MISLnet [5] results are simulated only for image resolution 64×64 . From Table 2, it can be deduced that RF-CNN achieves maximum detection accuracies in most of the cases, for both image resolutions 32×32 and 64×64 , as compared to state-of-the-art detectors. It should be noted that results obtained using ERT classifier are better than results obtained using softmax classifier in most of test case scenarios. The motivation for using softmax classifier is to facilitate comparison of RF-CNN results and MFR-CNN [10] results using same classifier. Moreover, it can be analyzed from Table 2 that RF-CNN with softmax classifier performs better than MFR-CNN [10] in majority of test cases.

Further, to show the efficacy of using F-layer in proposed RF-CNN, the average accuracies are determined for our proposed CNN (Figure 1) with F-layer and without F-layer. To this end, features of compressed ($Q = 70$) images of resolution S ($S = 64, 32$) manipulated with operations listed in Table 1 using highlighted parameters are determined and classified using softmax classifier. Results in Table 3 signify that proposed network with F-layer exhibits significant performance improvement for low-resolution images as compared to the proposed network without F-layer.

To validate the results shown in Table 2 and to obtain an insight about the effectiveness of trained two-class classifiers to classify various manipulation types, the mean activation maps [1, 36] of randomly selected 50 images of resolution 64×64 for both "Conv Layer 1" and "Conv Layer 2" have been analyzed. The 50 images are manipulated with each of seven different manipulation types using blue highlighted parameters shown in Table 1 and post-JPEG compressed with quality

Table 2. Single Manipulation Detection for Images of Resolution 32×32 and 64×64 under Compressed State

S	Q	Method/ Classifier	Org vs												
			Avgf (3 × 3)	Avgf (5 × 5)	Gauf (3 × 3)	Gauf (5 × 5)	Medf (3 × 3)	Medf (5 × 5)	Res (1.2)	Res (1.5)	Wiene rf(3 × 3)	Wiene rf(5 × 5)	Rot (1°)	Rot (3°)	Lapf (0.2)
64 × 64	70	RF-CNN/ Softmax	0.9951	0.9955	0.9079	0.9410	0.9889	0.9958	0.9951	0.9932	0.7752	0.8738	0.9815	0.9961	0.9908
		RF-CNN/ ERT	0.9980	0.9982	0.9359	0.9588	0.9932	0.9987	0.9994	0.9993	0.8365	0.9244	0.9913	0.9985	0.9964
		MFR-CNN	0.9986	0.9541	0.8961	0.9293	0.8957	0.9691	0.7539	0.8354	0.8433	0.9024	0.8776	0.7060	0.9962
		SRM	0.9964	0.9985	0.9295	0.9524	0.9493	0.9904	0.8371	0.9060	0.9572	0.9893	0.9586	0.9876	0.9978
		GLF	0.9723	0.9847	0.9380	0.9588	0.8928	0.9529	0.8399	0.9105	0.8491	0.9265	0.8098	0.8983	0.9912
		MISlnet/ Softmax	0.9978	0.9986	0.9078	0.9378	0.9874	0.9968	0.9397	0.9636	0.9606	0.9773	0.9904	0.9986	0.9917
		MISlnet/ ERT	0.9978	0.9989	0.9105	0.9401	0.9879	0.9968	0.9415	0.9630	0.9610	0.9781	0.9903	0.9986	0.9955
	30	RF-CNN/ Softmax	0.9808	0.9925	0.8372	0.9004	0.9456	0.9886	0.7423	0.8271	0.7079	0.7870	0.9610	0.9954	0.9883
		RF-CNN/ ERT	0.9943	0.9971	0.8914	0.9279	0.9729	0.9932	0.8188	0.8818	0.7669	0.8605	0.9837	0.9974	0.9971
		MFR-CNN	0.9408	0.9206	0.8188	0.8998	0.8075	0.8882	0.7609	0.8246	0.7484	0.8327	0.6913	0.6262	0.9759
32 × 32		SRM	0.9848	0.9939	0.8745	0.9184	0.8908	0.9736	0.7672	0.8416	0.8635	0.9617	0.9169	0.9807	0.9944
		GLF	0.9521	0.9804	0.9008	0.9330	0.8378	0.9284	0.7970	0.8731	0.7871	0.8886	0.7551	0.8629	0.9892
		MISlnet/ Softmax	0.9930	0.9972	0.8170	0.8695	0.9635	0.9906	0.8837	0.8791	0.8606	0.9705	0.9838	0.9939	0.9865
		MISlnet/ ERT	0.9933	0.9973	0.8163	0.8662	0.9639	0.9916	0.8830	0.8804	0.8607	0.9703	0.9844	0.9938	0.9916
	70	RF-CNN/ Softmax	0.9930	0.9958	0.8713	0.8907	0.9892	0.9971	0.9969	0.9971	0.8644	0.9244	0.9371	0.9967	0.9954
		RF-CNN/ ERT	0.9968	0.9974	0.8827	0.9173	0.9914	0.9980	0.9906	0.8741	0.9062	0.9555	0.9616	0.9987	0.9985
		MFR-CNN	0.9598	0.9759	0.8275	0.9538	0.9228	0.9684	0.6420	0.7441	0.7831	0.8397	0.5304	0.7640	0.9517
		SRM	0.9916	0.9958	0.8842	0.9218	0.9283	0.9870	0.7733	0.8480	0.9415	0.9816	0.8206	0.9455	0.9905
		GLF	0.9674	0.9828	0.8975	0.9314	0.8870	0.9486	0.7808	0.8640	0.8505	0.9277	0.6692	0.8790	0.9847
	30	RF-CNN/ Softmax	0.9964	0.9921	0.7843	0.8473	0.9577	0.9878	0.6442	0.7352	0.7478	0.8989	0.8738	0.9897	0.9889
64 × 64		RF-CNN/ ERT	0.9993	0.9943	0.8108	0.8552	0.9676	0.9936	0.9678	0.8113	0.8058	0.9277	0.9179	0.9926	0.9958
		MFR-CNN	0.9290	0.9517	0.7807	0.8322	0.7255	0.9421	0.6567	0.7409	0.6948	0.7807	0.5176	0.8058	0.9317
		SRM	0.9679	0.9878	0.8150	0.8685	0.8742	0.9692	0.7034	0.7815	0.8440	0.9468	0.7577	0.9039	0.9796
		GLF	0.9403	0.9748	0.8390	0.8861	0.8226	0.9288	0.7203	0.8081	0.7770	0.8904	0.6398	0.8251	0.9771

S and Q denote image resolution and quality factor of compression, respectively.

Table 3. Comparison of Average Accuracies Obtained for Two-class Classification with and without F-layer in RF-CNN

S	Method	Avgf (5×5)	Gauf (5×5)	Medf (5×5)	Res (1.5)	Wienerf (5×5)	Rot (3°)	Lapf (0.2)	Average
32 × 32	With F-layer	0.9958	0.8907	0.9971	0.9971	0.9244	0.9967	0.9954	0.9710
	Without F-layer	0.9967	0.8979	0.9964	0.8346	0.9480	0.9949	0.9908	0.9513
64 × 64	With F-layer	0.9955	0.9410	0.9958	0.9932	0.8738	0.9961	0.9908	0.9695
	Without F-layer	0.9977	0.9360	0.9918	0.8582	0.9199	0.9811	0.9847	0.9528

factor $Q = 70$. Further, using corresponding trained classifiers, the mean activation maps for 50 images, in both original and tampered form, are generated for randomly selected filter each from “Conv Layer 1” and “Conv Layer 2.” The activation map pairs for original and tampered images using “Conv Layer 1” filter are shown in columns 1 and 2 of Figure 5, respectively. Similarly, the activation map pairs for original and tampered image classes using “Conv Layer 2” filter are shown in columns 3 and 4 of Figure 5, respectively. Colorbar shown in rightmost column of Figure 5 shows color change from yellow to dark red to represent decreasing activation values.

As shown in row 1 of Figure 5, the mean activation maps for randomly selected 50 original and 50 tampered images, obtained using trained classifier for classification scenario “Org+jpeg 70 vs Avgf+jpeg70,” show that the activation map of “Conv Layer 1” filter for original images is maximally activated in comparison with activation map of average filtered images. Further, the activation map of “Conv Layer 2” filter for original images is activated in low-frequency regions whereas activation map of average filtered images is activated in high-frequency regions. These significant changes in activation maps for two classes of classification ensure almost perfect classification performance (accuracy=99.55%) as shown in Table 2. Similarly, for other manipulation types, i.e., median filtering (Medf), Laplacian filtering (Lapf), Gaussian filtering (Gauf), and rotation (Rot), the mean activation maps for original and tampered classes justify good classification performance of tampered images as compared to original images. However, for manipulation types rescaling (Res) and Wiener filtering (Wienerf), similar maximally activated regions can be observed in activation maps of original and tampered images. This results in relatively less classification accuracies for detection of rescaling and Wiener filtering operations.

7.2 Multiple Manipulation Detection

Multiple manipulation detectors play an important role in the field of image forensics, as they can be used in the preliminary phase of image processing history identification process chain to determine whether the candidate image has been tampered. Further, such detectors can be used to determine type of forgery used to tamper the image. Therefore, our proposed detector RF-CNN performance is evaluated for multi-class classification among eight different classes of classification. These eight classes of classification correspond to average filtering with window size 5, median filtering with window size 5, Wiener filtering with window size 5, Gaussian filtering with standard deviation $\sigma = 1.1$ and window size 5, Laplacian filtering with shape parameter $\alpha = 0.2$, image rotation with 3° followed by cropping back to original size, rescaling with factor 1.5 followed by cropping back to original size and original images. All images of eight different classes are JPEG compressed with quality factor $Q = 70$ and classification results for images of resolution $S = 32, 64, 128, 256$ are tabulated in Table 4–Table 7, respectively.

As manipulation detection in small resolution images is a challenging task, results of RF-CNN for small resolution images are compared with results of GLF [9], SRM_(T1,O3,q*) [34], MFR-CNN

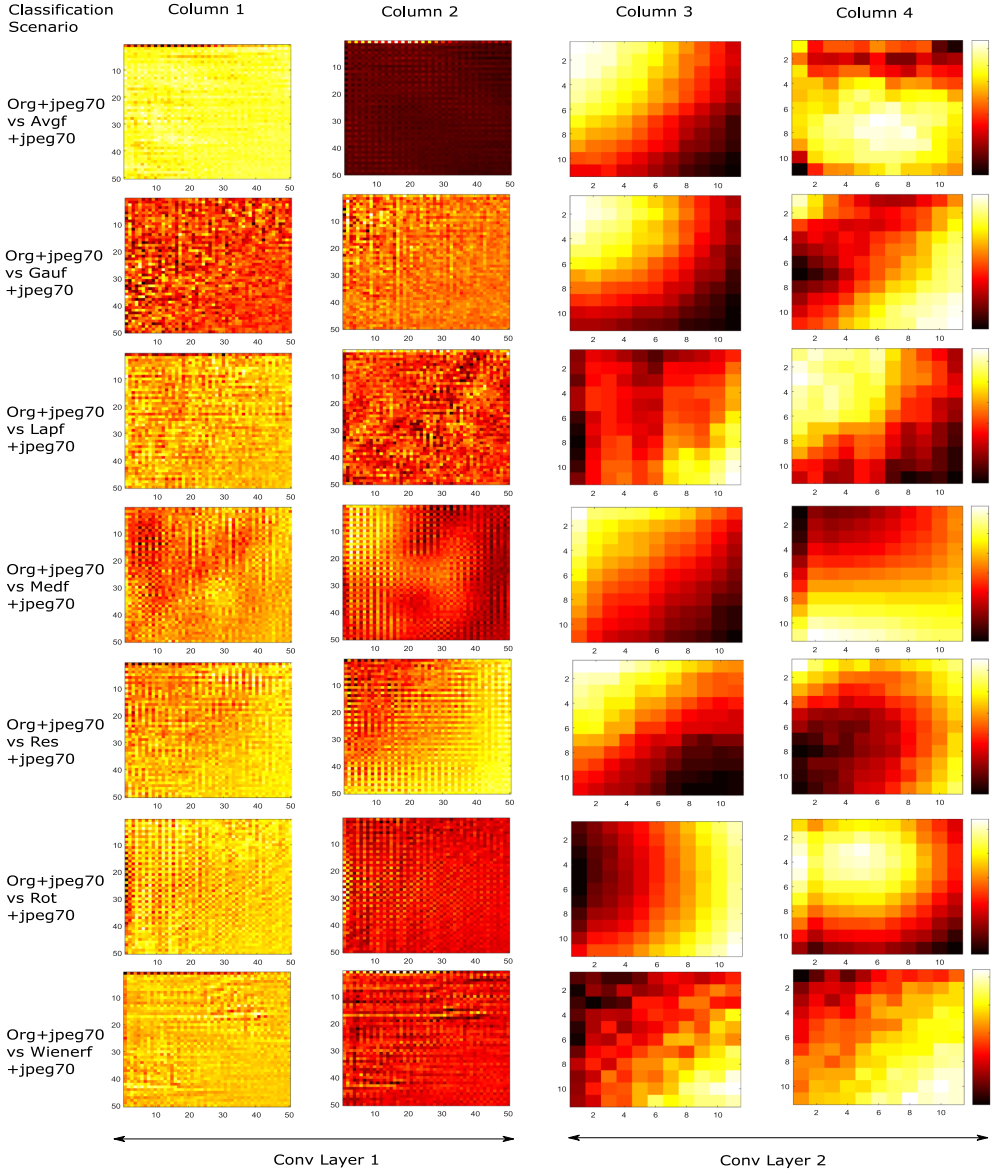


Fig. 5. Mean activation map pairs for original and tampered image class using filter of “Conv Layer 1” are shown in column 1 and column 2, respectively. Mean activation map pairs for original and tampered image class using filter of “Conv Layer 2” are shown in column 3 and column 4, respectively. Different classification scenarios are shown in left-most column from top to bottom.

[10], and MISLnet [5] detectors, which are shown in Tables 4 and 5 for image resolutions 32×32 and 64×64 , respectively. It should be noted that results for GLF [9], $\text{SRM}_{(T1, O3, q^*)}$ [34], and MFR-CNN [10] are simulated for $S = 32, 64$, whereas MISLnet [5] results are simulated only for $S = 64$.

Results reveal the efficacy of proposed detector RF-CNN to detect different manipulations as compared to state-of-the-art detectors, specifically when classified using ERT classifiers. It ensures comparable or better performance of proposed detector encompassed with the positive aspects of

Table 4. Comparison of RF-CNN Eight Class Classification Results with State-of-the-art Detectors to Identify Image Manipulation Type for Images of Size 32×32 Compressed with $Q = 70$

Actual/ Predicted	RF-CNN/Softmax/Average accuracy=0.8808								RF-CNN/ERT/Average accuracy=0.8945							
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf
Avgf	0.9897	0.0000	0.0017	0.0083	0.0000	0.0000	0.0000	0.0002	0.9924	0.0002	0.0031	0.0036	0.0000	0.0000	0.0000	0.0007
Gauf	0.0021	0.7168	0.0021	0.0024	0.0029	0.2393	0.0002	0.0341	0.0005	0.7957	0.0005	0.0010	0.0205	0.1645	0.0000	0.0174
Lapf	0.0100	0.0000	0.9509	0.0329	0.0002	0.0000	0.0014	0.0045	0.0010	0.0000	0.9928	0.0036	0.0000	0.0000	0.0007	0.0019
Medf	0.0038	0.0000	0.0029	0.9931	0.0000	0.0000	0.0002	0.0000	0.0024	0.0002	0.0064	0.9888	0.0005	0.0005	0.0000	0.0012
Org	0.0017	0.0391	0.0019	0.0010	0.7199	0.1928	0.0029	0.0408	0.0002	0.0672	0.0014	0.0000	0.7588	0.1404	0.0014	0.0305
Res	0.0029	0.1802	0.0024	0.0017	0.0412	0.7290	0.0005	0.0422	0.0005	0.2281	0.0012	0.0002	0.0501	0.6923	0.0010	0.0267
Rot	0.0012	0.0000	0.0010	0.0050	0.0005	0.0000	0.9912	0.0012	0.0007	0.0007	0.0021	0.0017	0.0000	0.0005	0.9931	0.0012
Wienerf	0.0052	0.0079	0.0038	0.0038	0.0150	0.0076	0.0007	0.9559	0.0019	0.0155	0.0021	0.0026	0.0217	0.0122	0.0019	0.9421
Actual/ Predicted	MFR-CNN/Softmax/Average accuracy=0.8807								GLF/Average Accuracy=0.8065							
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf
Avgf	0.9004	0.0000	0.0224	0.0162	0.0000	0.0033	0.0005	0.0572	0.8951	0.0333	0.0006	0.0035	0.0039	0.0137	0.0012	0.0486
Gauf	0.0000	0.9867	0.0105	0.0000	0.0000	0.0026	0.0000	0.0002	0.0086	0.8255	0.0026	0.0051	0.0041	0.1337	0.0005	0.0199
Lapf	0.0000	0.0000	0.9924	0.0014	0.0007	0.0007	0.0043	0.0005	0.0000	0.0000	0.9821	0.0000	0.0081	0.0092	0.0007	0.0000
Medf	0.0033	0.0002	0.0329	0.9235	0.0010	0.0026	0.0198	0.0167	0.0021	0.0230	0.0010	0.8442	0.0264	0.0343	0.0102	0.0588
Org	0.0000	0.0598	0.0086	0.0002	0.6827	0.2138	0.0103	0.0246	0.0011	0.0553	0.0152	0.0191	0.6905	0.1264	0.0591	0.0333
Res	0.0007	0.9940	0.0074	0.0005	0.0706	0.8036	0.0045	0.0133	0.0043	0.2242	0.0077	0.0150	0.0433	0.6498	0.0255	0.0303
Rot	0.0000	0.0005	0.0143	0.0021	0.0174	0.0067	0.9571	0.0019	0.0021	0.0055	0.0044	0.0184	0.1186	0.0637	0.7703	0.0171
Wienerf	0.0198	0.0215	0.0286	0.0067	0.0341	0.0858	0.0048	0.7988	0.0237	0.0523	0.0018	0.0456	0.0399	0.0359	0.0064	0.7944
Actual/ Predicted	SRM _(T1,O3,q3) /Average accuracy=0.8705															
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien erf
Avgf	0.9566	0.0010	0.0029	0.0033	0.0000	0.0031	0.0000	0.0331	0.0000	0.0031	0.0000	0.0000	0.0000	0.0000	0.0000	0.0331
Gauf	0.0007	0.8336	0.0114	0.0026	0.0043	0.1428	0.0005	0.0041	0.0043	0.1428	0.0005	0.0005	0.0005	0.0005	0.0005	0.0041
Lapf	0.0000	0.0002	0.9890	0.0005	0.0017	0.0069	0.0014	0.0002	0.0017	0.0069	0.0014	0.0002	0.0002	0.0002	0.0002	0.0002
Medf	0.0041	0.0062	0.0038	0.9528	0.0038	0.0105	0.0038	0.0150	0.0038	0.0105	0.0038	0.0150	0.0038	0.0105	0.0038	0.0150
Org	0.0002	0.0601	0.0150	0.0038	0.7194	0.1628	0.0358	0.0029	0.0002	0.0601	0.0150	0.0038	0.0358	0.1628	0.0358	0.0029
Res	0.0005	0.2327	0.0181	0.0067	0.0544	0.6625	0.0195	0.0057	0.0005	0.2327	0.0181	0.0067	0.0544	0.6625	0.0195	0.0057
Rot	0.0012	0.0021	0.0036	0.0029	0.0477	0.0308	0.9099	0.0019	0.0012	0.0021	0.0036	0.0029	0.0477	0.0308	0.9099	0.0019
Wienerf	0.0148	0.0110	0.0045	0.0119	0.0036	0.0119	0.0024	0.9399	0.0148	0.0110	0.0045	0.0119	0.0036	0.0119	0.0024	0.9399

Table 5. Comparison of RF-CNN Eight Class Classification Results with State-of-the-art Detectors to Identify Image Manipulation Type for Images of Size 64×64 Compressed with $Q = 70$

Actual/ Predicted	RF-CNN/Softmax/Average accuracy=0.8653										RF-CNN/ERT/Average accuracy=0.9089									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9883	0.0021	0.0043	0.0021	0.0002	0.0002	0.0007	0.0019			0.9959	0.0002	0.0005	0.0012	0.0000	0.0002	0.0000	0.0000	0.0019	
Gauf	0.0000	0.9714	0.0014	0.0000	0.0000	0.0000	0.0000	0.0033			0.0002	0.8679	0.0012	0.0005	0.0081	0.1092	0.0000	0.0000	0.0129	
Lapf	0.0000	0.0000	0.9816	0.0012	0.0057	0.0007	0.0019	0.0041			0.0000	0.0000	0.9981	0.0012	0.0002	0.0000	0.0000	0.0002	0.0002	
Medf	0.0026	0.0050	0.0112	0.9716	0.0005	0.0005	0.0031	0.0055			0.0024	0.0005	0.0024	0.9926	0.0000	0.0000	0.0014	0.0007	0.0071	
Org	0.0000	0.0965	0.0021	0.0000	0.7125	0.1142	0.0024	0.0722			0.0000	0.0458	0.0010	0.0002	0.7893	0.0868	0.0019	0.0751		
Res	0.0000	0.4424	0.0014	0.0000	0.0122	0.5240	0.0007	0.0193			0.0000	0.2043	0.0005	0.0007	0.0303	0.7344	0.0012	0.0286		
Rot	0.0000	0.0000	0.0041	0.0000	0.0064	0.0024	0.9824	0.0048			0.0000	0.0002	0.0014	0.0017	0.0014	0.0007	0.9924	0.0021		
Wienerf	0.0010	0.0765	0.0029	0.0000	0.0844	0.0415	0.0036	0.7902			0.0005	0.0257	0.0029	0.0014	0.0462	0.0217	0.0007	0.9008		
MFR-CNN/Softmax/Average accuracy=0.8790																				
Actual/ Predicted	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9733	0.0005	0.0026	0.0162	0.0000	0.0010	0.0002	0.0062			0.9242	0.0282	0.0001	0.0049	0.0044	0.0122	0.0007	0.0252		
Gauf	0.0010	0.6389	0.0036	0.0012	0.0052	0.3251	0.0005	0.0246			0.0130	0.8651	0.0006	0.0033	0.0026	0.0935	0.0002	0.0217		
Lapf	0.0007	0.0000	0.9952	0.0024	0.0002	0.0000	0.0002	0.0012			0.0000	0.0003	0.9901	0.0001	0.0066	0.0024	0.0003	0.0001		
Medf	0.0026	0.0002	0.0072	0.9657	0.0017	0.0007	0.0038	0.0181			0.0031	0.0189	0.0005	0.8457	0.0212	0.0315	0.0064	0.0727		
Org	0.0000	0.0210	0.0060	0.0012	0.8434	0.1042	0.0045	0.0198			0.0023	0.0308	0.0099	0.0225	0.7523	0.0809	0.0541	0.0473		
Res	0.0000	0.0954	0.0029	0.0000	0.0906	0.7795	0.0012	0.0305			0.0077	0.1573	0.0018	0.0197	0.0276	0.7381	0.0179	0.0297		
Rot	0.0000	0.0000	0.0043	0.0138	0.0188	0.0005	0.9600	0.0026			0.0030	0.0036	0.0009	0.0162	0.1082	0.0452	0.8119	0.0110		
Wienerf	0.0076	0.0067	0.0069	0.0136	0.0701	0.0176	0.0019	0.8756			0.0149	0.0370	0.0014	0.0696	0.0392	0.0302	0.0055	0.8021		
MISNet/Softmax/Average accuracy=0.9541																				
Actual/ Predicted	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9884	0.0000	0.0006	0.0013	0.0000	0.0000	0.0002	0.0096			0.9878	0.0000	0.0002	0.0015	0.0000	0.0000	0.0002	0.0103		
Gauf	0.0000	0.9640	0.0112	0.0004	0.0086	0.0112	0.0002	0.0043			0.0000	0.9412	0.0028	0.0004	0.0279	0.0217	0.0004	0.0056		
Lapf	0.0000	0.0004	0.9989	0.0000	0.0004	0.0004	0.0000	0.0000			0.0000	0.0015	0.9944	0.0000	0.0036	0.0006	0.0000	0.0000		
Medf	0.0021	0.0037	0.0004	0.9843	0.0017	0.0011	0.0007	0.0060			0.0007	0.0032	0.0002	0.9852	0.0022	0.0009	0.0006	0.0059		
Org	0.0002	0.1081	0.0174	0.0024	0.8203	0.0433	0.0019	0.0064			0.0000	0.0804	0.0075	0.9019	0.8559	0.0476	0.0011	0.0066		
Res	0.0000	0.0422	0.0127	0.0007	0.0272	0.9161	0.0007	0.0004			0.0000	0.0330	0.0039	0.0007	0.0371	0.9245	0.0002	0.0006		
Rot	0.0004	0.0006	0.0009	0.0004	0.0017	0.0000	0.9959	0.0002			0.0004	0.0006	0.0007	0.0006	0.0024	0.0000	0.9949	0.0004		
Wienerf	0.0071	0.0137	0.0011	0.0066	0.0041	0.0017	0.0004	0.9653			0.0047	0.0109	0.0004	0.0058	0.0052	0.0021	0.0004	0.9706		
SRM _(T1, O3, q5) /Average accuracy=0.9231																				
Actual/ Predicted	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9859	0.0005	0.0012	0.0029	0.0000	0.0005	0.0002	0.0088			0.9859	0.0005	0.0012	0.0029	0.0000	0.0005	0.0002	0.0088		
Gauf	0.0002	0.8827	0.0041	0.0024	0.0000	0.0000	0.0000	0.0026			0.0002	0.8827	0.0041	0.0024	0.0000	0.0000	0.0000	0.0026		
Lapf	0.0000	0.0000	0.9959	0.0005	0.0000	0.0000	0.0002	0.0000			0.0000	0.0000	0.9959	0.0005	0.0000	0.0000	0.0002	0.0000		
Medf	0.0021	0.0021	0.0029	0.9766	0.0021	0.0041	0.0029	0.0072			0.0021	0.0021	0.0029	0.9766	0.0021	0.0041	0.0029	0.0072		
Org	0.0000	0.0386	0.0041	0.0043	0.8327	0.1082	0.0086	0.0036			0.0000	0.0386	0.0041	0.0043	0.8327	0.1082	0.0086	0.0036		
Res	0.0000	0.1788	0.0060	0.0033	0.0458	0.7588	0.0050	0.0024			0.0000	0.1788	0.0060	0.0033	0.0458	0.7588	0.0050	0.0024		
Rot	0.0002	0.0000	0.0019	0.0019	0.0000	0.0033	0.9821	0.0005			0.0002	0.0000	0.0019	0.0019	0.0000	0.0033	0.9821	0.0005		
Wienerf	0.0057	0.0048	0.0019	0.0079	0.0024	0.0057	0.0017	0.9700			0.0057	0.0048	0.0019	0.0079	0.0024	0.0057	0.0017	0.9700		

Table 6. RF-CNN Confusion Matrices to Identify Multiple Image Manipulations for Images of Size 128×128 using Softmax and ERT Classifiers

Actual/ Predicted	Softmax/Average accuracy=0.8902										Extremely randomized trees/Average accuracy=0.9239									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9974	0.0002	0.0019	0.0002	0.0000	0.0000	0.0000	0.0000	0.0002		0.9969	0.0002	0.0010	0.0012	0.0000	0.0002	0.0000	0.0000	0.0005	
Gauf	0.0002	0.9228	0.0012	0.0007	0.0036	0.0613	0.0002	0.0100	0.0100		0.0007	0.9619	0.0002	0.0012	0.0055	0.0129	0.0005	0.0172	0.0005	
Lapf	0.0000	0.0000	0.9974	0.0010	0.0007	0.0000	0.0002	0.0007	0.0007		0.0000	0.0000	0.9976	0.0010	0.0000	0.0000	0.0002	0.0012	0.0012	
Medf	0.0050	0.0002	0.0043	0.9876	0.0002	0.0000	0.0002	0.0024	0.0024		0.0021	0.0002	0.0014	0.9938	0.0000	0.0002	0.0005	0.0005	0.0017	
Org	0.0002	0.0303	0.0038	0.0000	0.7878	0.0791	0.0021	0.0965	0.0965		0.0000	0.0403	0.0021	0.0002	0.7609	0.0837	0.0026	0.1104	0.1104	
Res	0.0005	0.2589	0.0026	0.0002	0.0131	0.7001	0.0000	0.0246	0.0246		0.0007	0.0353	0.0019	0.0005	0.0238	0.8777	0.0021	0.0579	0.0579	
Rot	0.0000	0.0007	0.0036	0.0005	0.0095	0.0038	0.9719	0.0100	0.0100		0.0000	0.0010	0.0012	0.0005	0.0081	0.0043	0.9805	0.0045	0.0045	
Wienerf	0.0019	0.0710	0.0117	0.0014	0.1206	0.0355	0.0012	0.7566	0.7566		0.0010	0.0415	0.0048	0.0048	0.0844	0.0381	0.0031	0.8224	0.8224	

Table 7. RF-CNN Confusion Matrices to Identify Multiple Image Manipulations for Images of Size 256×256 using Softmax and ERT Classifiers

Actual/ Predicted	Softmax/Average accuracy=0.8936										Extremely randomized trees/Average accuracy=0.9200									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9955	0.0007	0.0014	0.0019	0.0000	0.0000	0.0005	0.0000	0.0000		0.9969	0.0000	0.0005	0.0014	0.0005	0.0000	0.0002	0.0005	0.0005	
Gauf	0.0002	0.9428	0.0010	0.0005	0.0042	0.0000	0.0129	0.0007	0.0007		0.0005	0.9187	0.0005	0.0005	0.0646	0.0000	0.0131	0.0021	0.0021	
Lapf	0.0000	0.0000	0.9883	0.0005	0.0000	0.0005	0.0033	0.0074	0.0074		0.0000	0.0000	0.9986	0.0000	0.0000	0.0002	0.0007	0.0005	0.0005	
Medf	0.0026	0.0005	0.0019	0.9912	0.0002	0.0002	0.0029	0.0005	0.0005		0.0002	0.0002	0.0012	0.9967	0.0002	0.0000	0.0010	0.0005	0.0005	
Org	0.0000	0.2198	0.0010	0.0010	0.7247	0.0002	0.0319	0.0215	0.0215		0.0000	0.1235	0.0007	0.0005	0.8477	0.0005	0.0198	0.0074	0.0074	
Res	0.0000	0.0005	0.0012	0.0000	0.0010	0.9578	0.0069	0.0327	0.0327		0.0000	0.0002	0.0019	0.0002	0.0000	0.9831	0.0055	0.0091	0.0091	
Rot	0.0007	0.0391	0.0026	0.0045	0.0315	0.0007	0.7788	0.1421	0.1421		0.0000	0.0312	0.0055	0.0029	0.0250	0.0012	0.8334	0.1009	0.1009	
Wienerf	0.0002	0.0212	0.0031	0.0012	0.0474	0.0002	0.1569	0.7697	0.7697		0.0000	0.0112	0.0033	0.0002	0.0427	0.0012	0.1564	0.7849	0.7849	

Table 8. Comparison of Average Accuracies Obtained for Eight Class Classification with and without F-layer in RF-CNN

S	With F-layer	Without F-layer
256	0.8936	0.7868
128	0.8902	0.9002
64	0.8653	0.8626
32	0.8808	0.8636

Table 9. Image Manipulation Types with All Considered Editing Parameters

Manipulation type	Parameters
Average filtering (Avgf)	$w_{size} = 3, 5, 7, 9$
Gaussian filtering (Gauf)	$w_{size} = 3, 5, 7, 9$
Laplacian filtering (Lapf)	$\alpha = 0.1, 0.2, \dots, 0.9$
Median filtering (Medf)	$w_{size} = 3, 5, 7, 9$
Rescaling operation (Res) using bilinear interpolation	Scale = 1.1, 1.2, $\dots$, 2.0
Rotation operation (Rot) followed by cropping to same size	$0.5^\circ, 1^\circ, 1.5^\circ, \dots, 5^\circ$
Wiener filtering (Wienerf)	$w_{size} = 3, 5, 7, 9$

machine learning-based detectors (small simulation time) and CNN-based detectors (high accuracy).

Additionally, average accuracies for multi-class classification of 32×32 and 64×64 resolution images compressed with $Q = 70$ are compared for RF-CNN with F-layer and RF-CNN without F-layer. The comparison results are tabulated in Table 8. Results in Table 8 signify that RF-CNN with F-layer exhibits better accuracies as compared to RF-CNN without F-layer, especially for low-resolution (32×32) and high-resolution images (256×256). For high-resolution images, the significant drop in performance of proposed network without F-layer is due to improper feature extraction of input residual images by two-layer network and result in improperly trained model. However, due to spatial invariance property of frequency domain features [15], the frequency domain transformation of image MFRs facilitates complete training of RF-CNN network for all resolution images down to 32×32 .

7.3 Multiple Manipulation Detection Using Random Parameters

To verify the robustness of proposed method against parameters used for editing operation, a set of random parameters is generated for multi-class classification. These parameters are randomly selected from parameters given in Table 9. The image editing operations for random parameter multiple manipulation detection are average filtering with window size $w_{size} = 9$, Gaussian filtering with window size $w_{size} = 9$ and standard deviation $\sigma = 2$ determined by $w_{size} = 2[(2 \times \sigma)] + 1$ (MATLAB imgaussfilt function), Laplacian filtering with shape parameter $\alpha = 0.6$, median filtering with window size $w_{size} = 3$, rescaling with factor 1.3, rotation with 0.5° , and Wiener filtering with window size $w_{size} = 9$. Confusion matrices for random parameters multiple manipulation detection using softmax and ERT classifiers are shown in Table 10 and Table 11 for image sizes $S = 32$ and 64, respectively. Results ensure the robustness of proposed detector RF-CNN against randomly selected parameters used to manipulate the images.

Table 10. RF-CNN Confusion Matrices to Identify Image Manipulations Performed with Random Parameters Selected from Table 9 for Images of Size 32×32

Actual/ Predicted	Softmax/Average accuracy=0.9153										Extremely randomized trees/Average accuracy=0.9345									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9807	0.0000	0.0136	0.0005	0.0019	0.0000	0.0000	0.0033			0.9924	0.0000	0.0038	0.0007	0.0000	0.0000	0.0000	0.0031		
Gauf	0.0005	0.9719	0.0045	0.0000	0.0041	0.0067	0.0012	0.0112			0.0010	0.9483	0.0010	0.0007	0.0002	0.0396	0.0014	0.0079		
Lapf	0.0010	0.0000	0.9838	0.0031	0.0079	0.0000	0.0120	0.0031			0.0021	0.0000	0.9921	0.0024	0.0012	0.0000	0.0010	0.0012		
Medf	0.0000	0.0033	0.0138	0.9638	0.0052	0.0005	0.0055	0.0079			0.0012	0.0041	0.0081	0.9716	0.0002	0.0029	0.0050	0.0069		
Org	0.0000	0.0002	0.0098	0.0000	0.9874	0.0000	0.0019	0.0007			0.0000	0.0000	0.0005	0.0000	0.9983	0.0000	0.0012	0.0000		
Res	0.0002	0.1703	0.0038	0.0048	0.0036	0.6203	0.1151	0.0729			0.0002	0.1151	0.0007	0.0048	0.0002	0.7883	0.0644	0.0262		
Rot	0.0000	0.0124	0.0031	0.0062	0.0041	0.0400	0.8713	0.0629			0.0002	0.0107	0.0017	0.0079	0.0002	0.0796	0.8634	0.0362		
Wienerf	0.0012	0.0083	0.0083	0.0057	0.0048	0.0038	0.0246	0.9433			0.0029	0.0081	0.0033	0.0081	0.0002	0.0167	0.0386	0.9220		

Classification is performed using softmax and ERT classifiers.

Table 11. RF-CNN Confusion Matrices to Identify Image Manipulations Performed with Random Parameters Selected from Table 9 for Images of Size 64×64

Actual/ Predicted	Softmax/Average accuracy=0.8595										Extremely randomized trees/Average accuracy=0.9097									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9824	0.0005	0.0157	0.0005	0.0000	0.0002	0.0000	0.0007			0.9964	0.0005	0.0014	0.0000	0.0000	0.0000	0.0000	0.0017		
Gauf	0.0012	0.9743	0.0079	0.0045	0.0107	0.0012	0.0000	0.0002			0.0002	0.9735	0.0000	0.0010	0.0074	0.0129	0.0000	0.0050		
Lapf	0.0010	0.0000	0.9945	0.0021	0.0012	0.0000	0.0005	0.0007			0.0002	0.0000	0.9986	0.0000	0.0002	0.0000	0.0002	0.0007		
Medf	0.0002	0.0012	0.0072	0.9864	0.0000	0.0012	0.0017	0.0021			0.0007	0.0024	0.0014	0.9843	0.0005	0.0029	0.0026	0.0055		
Org	0.0002	0.0501	0.0062	0.0098	0.6405	0.0906	0.0434	0.1592			0.0002	0.0272	0.0005	0.0048	0.7077	0.1299	0.0663	0.0634		
Res	0.0007	0.1037	0.0064	0.0143	0.1516	0.6400	0.0153	0.0679			0.0007	0.0658	0.0002	0.0067	0.0436	0.8384	0.0176	0.0269		
Rot	0.0010	0.0029	0.0103	0.0176	0.1030	0.0300	0.7857	0.0496			0.0007	0.0029	0.0002	0.0081	0.0551	0.0260	0.8858	0.0212		
Wienerf	0.0026	0.0272	0.0148	0.0181	0.0381	0.0122	0.0145	0.8725			0.0021	0.0098	0.0062	0.0062	0.0424	0.0215	0.0186	0.8932		

Classification is performed using softmax and ERT classifiers.

7.4 Time Complexity of CNN-based Methods

Various real-time image processing applications require the CNN-based classifiers to respond in constrained time. The increase in the hardware computational capability may speed up the computations but with the excessive hardware cost overhead. Additionally, it can be difficult to produce results in real-time for portable devices with low computational power [21]. Thus, the success of a forensic detector, especially for portable devices, critically depends upon the time taken by the detector to respond to a particular classification problem. In this respect, the time complexity (τ) of proposed RF-CNN along with state-of-the-art CNN-based detectors MFR-CNN [10] and MISLnet [5] is evaluated using the formula [21]

$$\tau = O\left(\sum_{l=1}^d f_{l-1} \cdot s_l^2 \cdot f_l \cdot o_l^2\right). \quad (5)$$

The above given formula determines the time complexity of all convolutional layers of the network. Here, d represents depth of the network (number of convolutional layers), f_l denotes number of filters in l th layer, s_l denotes spatial size of the filters of layer l and o_l represents the spatial size of output feature map of layer l .

Further, relative time complexity of MFR-CNN and MISLnet convolutional layers with respect to time complexity of RF-CNN convolutional layers is determined and found that MFR-CNN is 40.28 times slower than RF-CNN and MISLnet is 8.75 times slower than RF-CNN. Additionally, the training time elapsed by each of RF-CNN, MFR-CNN, and MISLnet, for eight class classification scenario, is also determined and found 1.03 hours, 10.88 hours, and 10.22 hours for RF-CNN, MFR-CNN, and MISLnet, respectively. The quantitative time elapsed for all methods is computed on Lenovo P310 workstation with Intel Xenon processor and 8 GB RAM without employing the GPU. The difference between theoretical time complexities and quantitative time values can be justified with the argument that theoretical time complexities are determined for convolutional layers only. It should be noted that the time complexities and quantitative time values are determined for RF-CNN, MFR-CNN, and MISLnet assuming input image of dimensions $64 \times 64 \times 1$.

The above analysis signifies that the proposed RF-CNN exhibits minimum time complexity as compared to MFR-CNN, MISLnet and is appropriate to be used for scenarios with constrained time budget.

7.5 Performance for Cross-database Images

To analyze the performance of RF-CNN along with state-of-the-art detectors for their capability to detect unseen tampered images, the trained models for image database Φ^S ($S = 32 \times 32$) are used to detect 1,488 compressed ($Q = 70$) original and 1,488 compressed tampered images of Dresden image database [19]. The testing results for single manipulation detection are obtained using softmax and presented in terms of receiver operating characteristic (ROC) curves in Figure 6. The results are determined for images, images being manipulated with seven different manipulation types with blue highlighted parameters given in Table 1. Furthermore, testing results for multi-class classification are obtained in terms of average testing accuracies and are shown in Table 12. It should be noted that for determination of eight class classification results, 1,488 images of post-JPEG compressed ($Q = 70$) original class images and each of seven different manipulation class images have been used. Multiclass classification results are determined for both image resolutions $S = 32 \times 32$, 64×64 , whereas single manipulation detection results are determined for low-resolution images, i.e., $S = 32 \times 32$.

From Figure 6, it can be deduced that the proposed RF-CNN detector exhibits almost perfect classification performance for all manipulation types with better performance for detection of

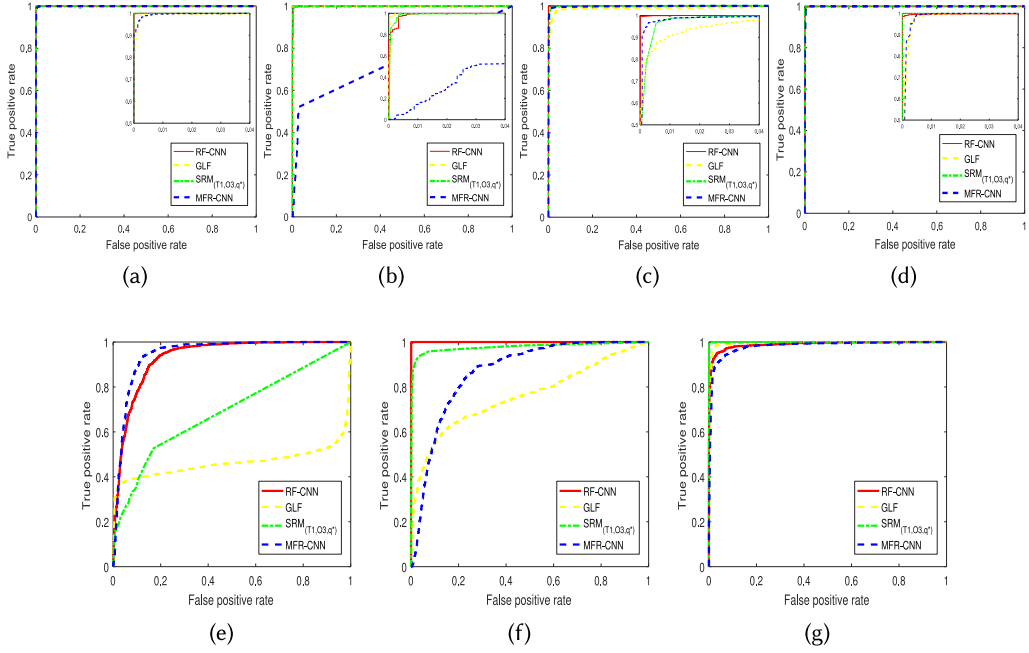


Fig. 6. ROC curves to represent testing accuracies for detection of 32×32 sized compressed ($Q = 70$) tampered images against corresponding compressed ($Q = 70$) original images of DID [19] database. Plots are shown for detection of (a) Average filtering, (b) Gaussian filtering, (c) Median filtering, (d) Laplacian filtering, (e) Rescaling, (f) Rotation, and (g) Wiener filtering.

Table 12. Cross-database Average Testing Accuracies for Eight Class Classification.

S	RF-CNN (Softmax)	RF-CNN (ERT)	MFR-CNN	SRM _(T1,O3,q*)	GLF	MISLnet (Softmax)	MISLnet (ERT)
32×32	0.8452	0.8371	0.8224	0.8448	0.7293	-	-
64×64	0.8164	0.8080	0.7571	0.6389	0.6866	0.8407	0.8418

S Represents Image Resolution.

median filtering and rotation operations as compared to state-of-the-art detectors. However, for other manipulation types, detector RF-CNN exhibits comparable performance with respect to performance of existing methods. Furthermore, the average testing accuracies for multiclass classification shown in Table 12 ensure the effectiveness of proposed RF-CNN to detect image manipulations in low-resolution ($S = 32 \times 32$) compressed images as compared to state-of-the-art detectors.

Recently, References [2] and [38] have shown real-world scenarios where there is a mismatch between the post-JPEG compression quality factor used in training and testing images. Therefore, the performance of proposed RF-CNN (using softmax) is also evaluated for the case when there is a mismatch between the post-JPEG compression quality factors of training and testing images. In this respect, different models are trained using the database Φ^S ($S = 32 \times 32$) for detection of two manipulation types: median filtering (window size $w_{size} = 5$) and rescaling (Scale=1.5). For each manipulation type, first model is trained using uncompressed (UnC) images and second model is trained using a mix of images compressed with two different quality factors $Q_{train} = 80, 90$. Further, the trained models are used to test 1,488 original images (DID database [19]) and 1,488 corresponding manipulated images compressed with different quality

Table 13. Cross-database Testing Accuracies (Using Softmax) under Mismatch between Post-JPEG Compression Quality Factors of Training (Q_{train}) and Testing (Q_{test}) Images

Q_{train}	Manipulation Type	Q_{test}				
		UnC	95	90	85	70
UnC	Medf	99.93	99.40	99.29	99.40	99.40
	Res	89.55	69.42	66.47	66.90	73.22
80,90	Medf	99.87	99.60	99.50	99.43	99.53
	Res	76.41	73.12	71.40	70.30	70.56

factors $Q_{test} = \text{UnC}, 95, 90, 85, 70$. The testing results are shown in Table 13. Results show that the models trained for detection of median filtering can consistently detect the same in both uncompressed and compressed (compressed with Q_{test}) images, irrespective of the mismatch between the post-JPEG compression quality factors of training and testing images. However, for rescaling operation, testing results are slightly degraded when there is an increase in mismatch between the compression quality factors of training and testing images.

7.6 Performance Improvement for Large Databases

In proposed work, we aim to achieve high-accuracy models for multiple image manipulation detection with significant reduction in training time and large dataset requirement. Thus, a database of 13,985 original images has been selected for training proposed RF-CNN network. However, the results for large database used in Reference [5] have also been tabulated in Table 14. It can be observed that with very large database size (133,336 training images), the average detection accuracies for eight class classification of 32×32 and 64×64 sized images compressed with quality factor $Q = 70$ are slightly increased from 89.45% to 90.06% and from 90.89% to 92.01%, respectively. It should be noted that ERT classifier is used for classification.

8 CONCLUSION

In this article, we proposed a novel multiple manipulation detector (RF-CNN). RF-CNN simultaneously extracts multiple forgery detection features by analyzing the frequency domain behavior of original and manipulated image MFRs. Experimental evidences have proved that different content-preserving manipulations leave distinguishable artifacts in frequency domain images (Figure 5). Frequency domain features are extracted using relatively small CNNs and classified using two different classifiers—softmax and extremely randomized trees (ERT). RF-CNN was compared against existing hand-crafted features-based detectors GLF [9], SRM_(T1, O3, q*) [34] and CNN-based detectors MFR-CNN [10], MISLnet [5]. The effectiveness of the proposed method was observed using rigorous experimentation results for two class and multiclass classification. For two class classification, our method achieves a maximum average validation accuracy for images of low resolution, i.e., 32×32 , compressed with quality factor $Q = 70, 30$. For multi-class classification, RF-CNN can detect the specific manipulation type among a pool of various operations with maximum average accuracy for smaller image resolutions (32×32) compressed with quality factor $Q = 70$. The advantages of the proposed RF-CNN as compared to existing detectors are listed below:

- (1) Frequency transformation converts RF-CNN into a generalized network to be adapted for images of all resolutions down to 32×32 .
- (2) Lower time complexity of RF-CNN as compared to existing CNN-based methods.

Table 14. RF-CNN Confusion Matrices to Identify Image Manipulation Type for Images of Size 32×32 and 64×64 Using ERT Classifier for Database Used in Reference [5]

Actual/ Predicted	32×32 /Average accuracy=0.9006										64×64 /Average accuracy=0.9201									
	Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf		Avgf	Gauf	Lapf	Medf	Org	Res	Rot	Wien	erf	
Avgf	0.9962	0.0000	0.0027	0.0011	0.0000	0.0000	0.0000	0.0000	0.0000		0.9994	0.0000	0.0004	0.0002	0.0000	0.0000	0.0000	0.0000	0.0000	
Gauf	0.0000	0.8082	0.0006	0.0002	0.0242	0.1594	0.0000	0.0074			0.0000	0.8546	0.0002	0.0002	0.0127	0.1250	0.0000	0.0074		
Lapf	0.0008	0.0000	0.9964	0.0015	0.0002	0.0000	0.0004	0.0008			0.0002	0.0000	0.9985	0.0002	0.0002	0.0000	0.0008	0.0002		
Medf	0.0008	0.0000	0.0021	0.9966	0.0000	0.0000	0.0000	0.0006			0.0002	0.0000	0.0011	0.9983	0.0000	0.0000	0.0002	0.0002		
Org	0.0000	0.1011	0.0013	0.0002	0.7451	0.1447	0.0002	0.0074			0.0000	0.0506	0.0008	0.0000	0.8055	0.1182	0.0013	0.0237		
Res	0.0000	0.2401	0.0002	0.0002	0.0691	0.6840	0.0002	0.0062			0.0002	0.1943	0.0004	0.0004	0.0517	0.7453	0.0000	0.0078		
Rot	0.0000	0.0000	0.0013	0.0002	0.0011	0.0002	0.9968	0.0004			0.0000	0.0000	0.0002	0.0002	0.0008	0.0004	0.9981	0.0004		
Wienerf	0.0000	0.0053	0.0025	0.0009	0.0061	0.0038	0.0004	0.9811			0.0008	0.0089	0.0028	0.0008	0.0189	0.0059	0.0006	0.9614		

- (3) Requirement of relatively small number of training images to properly train RF-CNN network.
- (4) Better universal detector to detect manipulation types in low-resolution compressed images.

However, for large resolution images, RF-CNN under-performs compared to existing state-of-the-art detectors. The proposed work thus provides a new direction for image manipulation detection in low-resolution images. Given recent reports [43], such capabilities will be crucially required. The proposed work can be further explored for development of video forensic detectors.

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